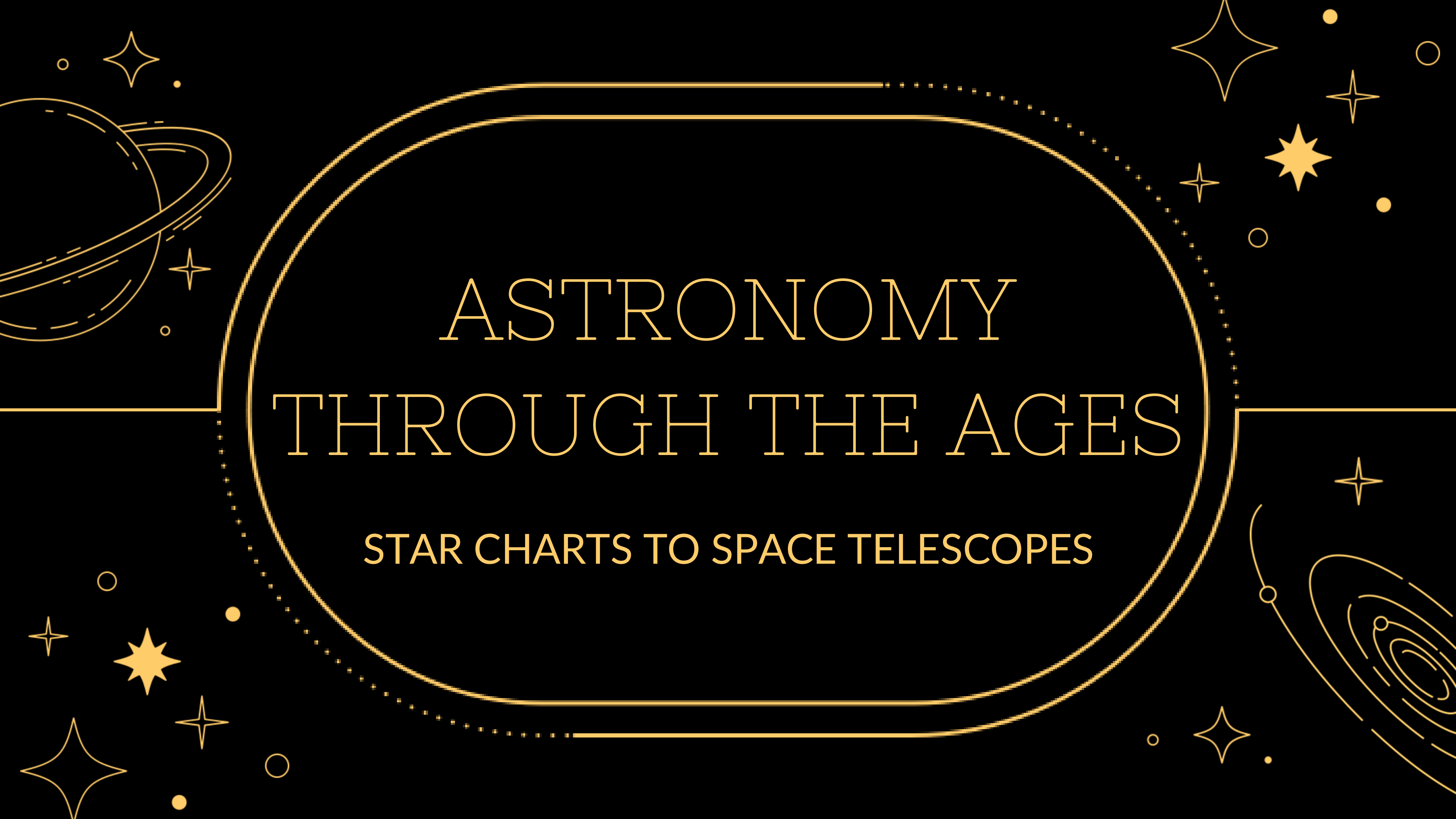




ASTRONOMY THROUGH THE AGES

STAR CHARTS TO SPACE TELESCOPES

Overview

Prehistory-1500

Ancient to Medieval times:

- Cave paintings
- Star Charts and catalogs
- Astrolabes
- Quadrants / Sextants

1600s-1800s

Scientific Revolution and
Enlightenment:

- Refracting Telescope
- Reflecting Telescope
- First photograph of the
moon

1900s - Present

Modern Innovations:

- Apollo Missions
- Hubble Telescope
- Kepler Space Telescope
- James Webb Telescope

Looking Up

Looking to the stars is a part of our humanity



My mom started taking me stargazing when I was small. I grew up on the stories of the constellations, and they've been my constant friends since.

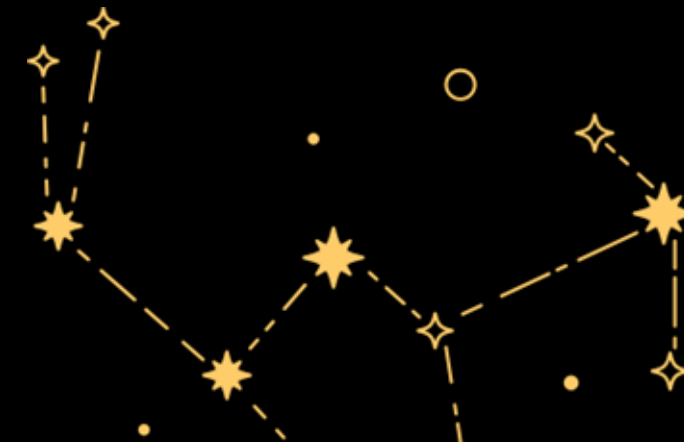


Is it any wonder that people have been looking to the stars for answers, perhaps since we gained the ability to see?

Cave Art and Star Charts



Researchers have found evidence that cave drawings previously assumed to just be animals, actually represent constellations. Based on the chemical makeup of the paint, humans could have been using stars to tell time more than 40,000 years ago



[article](#)



Mesopotamian Boundary Stone - 11th Century BC

Currently sitting in the British Museum, this limestone kudurru potentially depicts some of the Zodiac Constellations.

Before the Greeks

Babylonian Star Catalogues - 320-150 BC



Ancient cuneiform texts that systematically list stars, constellations, and planets, reflecting a sophisticated understanding of the night sky. astronomy.





DID YOU KNOW?



The western zodiac group of constellations was named by Ptolemy, a Greek-born Egyptian astronomer, mathematician, and geographer. It is based on the Sun's path in the sky throughout the year. Hence, these constellations differ in visibility and position during different seasons.



Hypatia's Astrolabe (c. 370–415 AD)

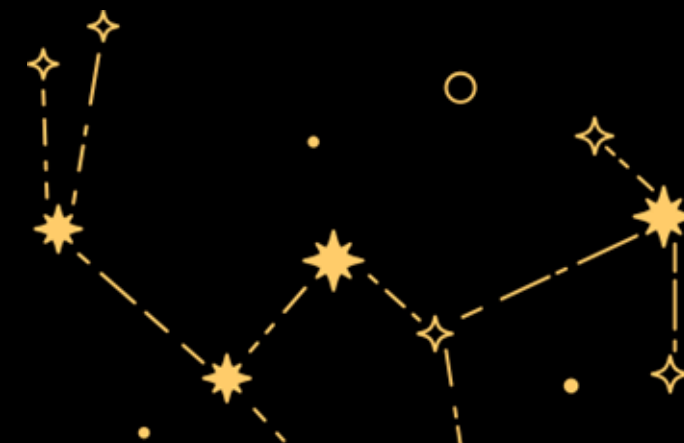
This astrolabe “was made in the 1220s CE for Abu-l-Fatih Musa, the nephew of Sultan Saladin, the famous conqueror of Jerusalem during the period of the Crusades.”

Based on inscription about 1227 AD



Although, Hypatia did not invent the astrolabe as it's origins trace to earlier times, she is credited for refining and improving the device and teaching its use to her pupils such as Synesius.

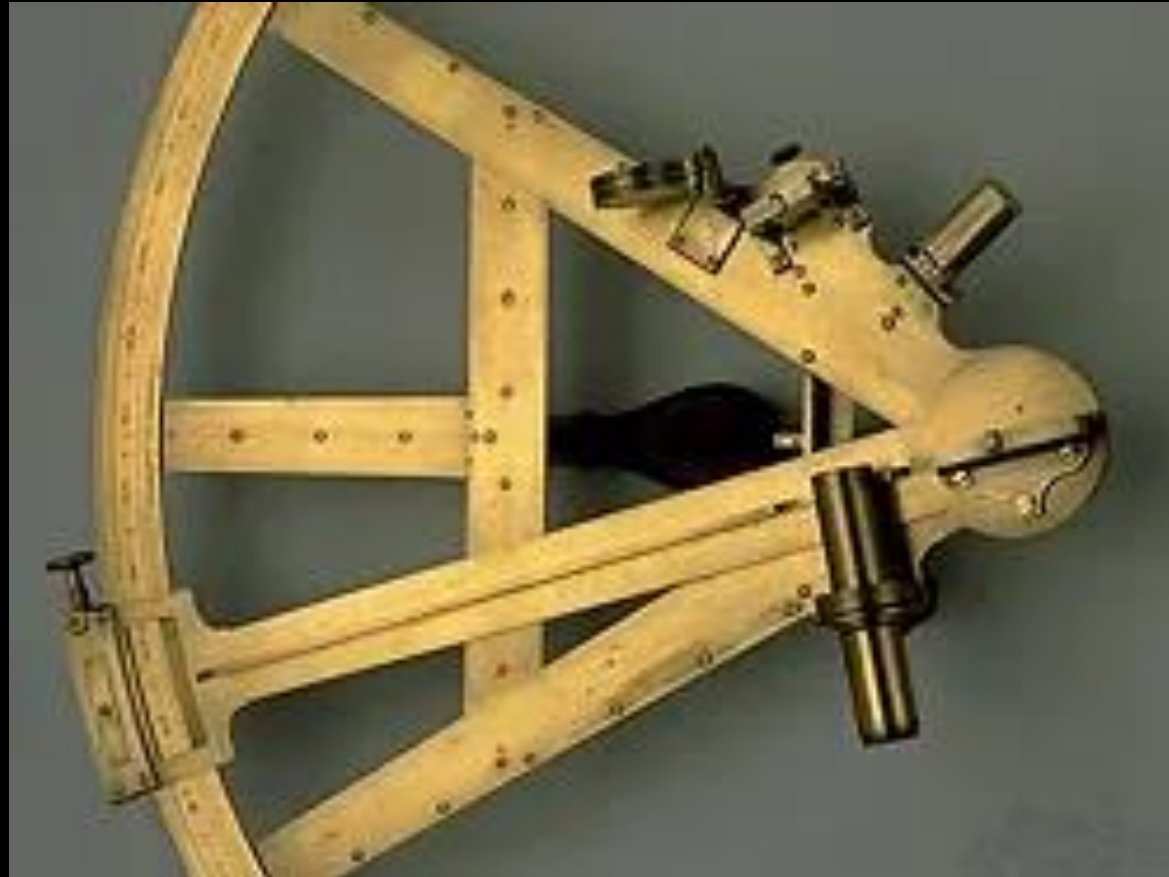
It is used to measure the altitude of stars or the sun. By aligning the rotating star map (called the rete) with the date and time, and sighting a celestial object using the alidade on the back, users could determine local time, latitude, or even make astrological predictions. It was like a multi-tool for medieval astronomers—used for navigation, timekeeping, and observing the heavens.



[images](#)



Islamic Contributions

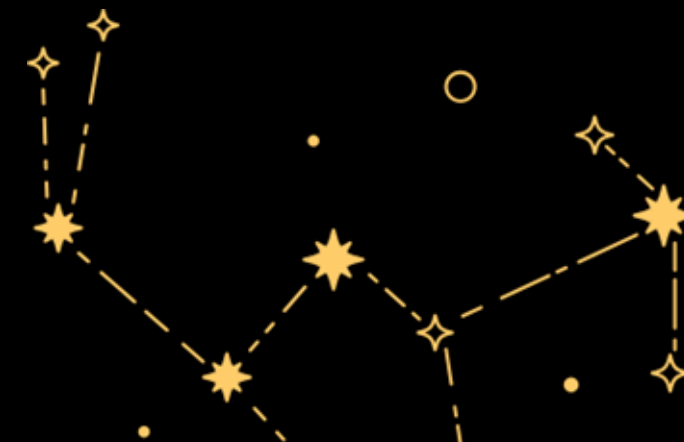


8th–13th AD — Islamic scholars preserved and improved ancient Greek astronomical texts.

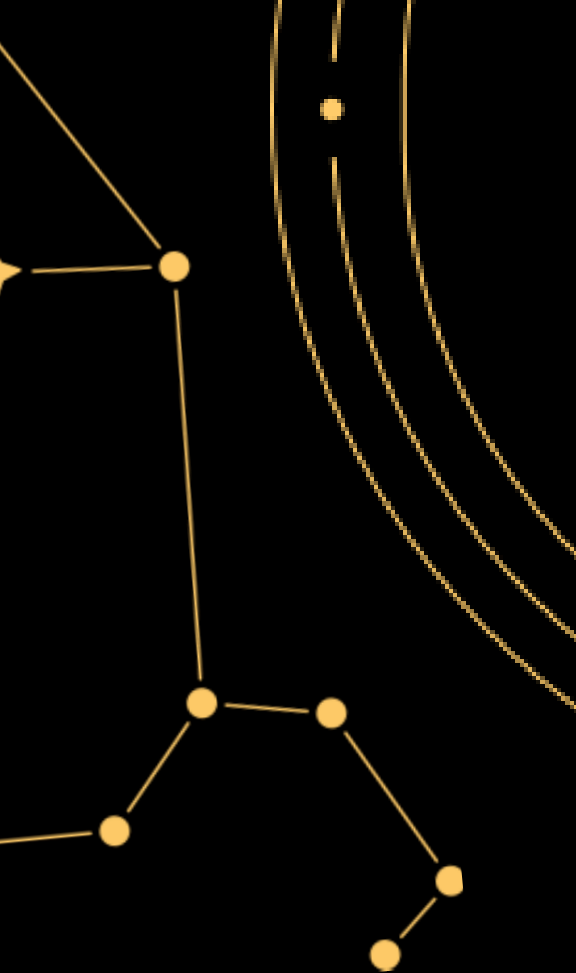
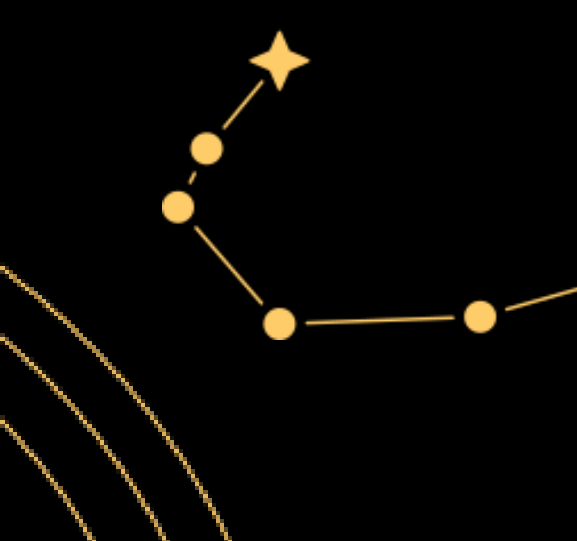
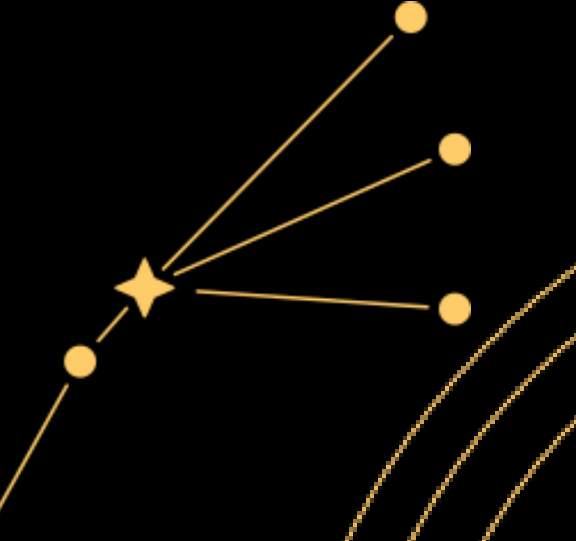
- Inventions: Quadrants, Sextants, refinement of astrolabe, zij star tables (e.g., Al-Zarqali).

~1250 AD — Maragha Observatory (Persia): Pioneered new observational techniques.

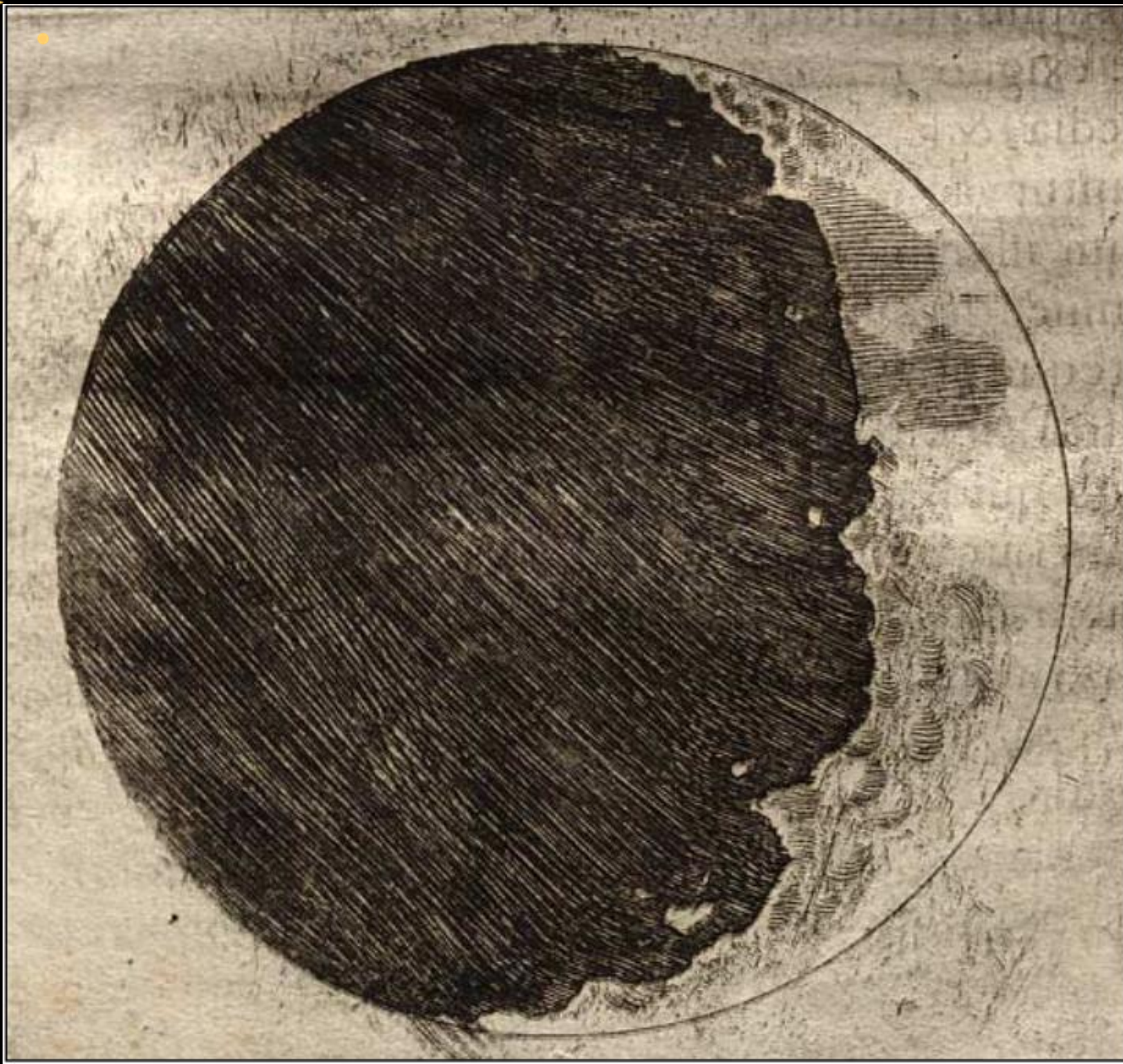
15th AD — Ulugh Beg's star catalog included some of the most accurate measurements pre-telescope.



sextant



SCIENTIFIC REVOLUTION & ENLIGHTENMENT



Sidereus Nuncius

“And we have an almost entirely similar sight on Earth, around sunrise, when the valleys are not yet bathed in light but the surrounding mountains facing the Sun are already seen shining with light. And just as the shadows of the earthly valleys are diminished as the Sun climbs higher, so those lunar spots lose their darkness as the luminous part grows.” - *Sidereus Nuncius* by Galileo Galilei (1610).



Galileo's Refracting Telescope

- 1 Observed craters and mountains on the Moon, proving it wasn't a perfect sphere.
- 2 Discovered moons orbiting Jupiter, showing that not everything revolves around the Earth.
- 3 Allowed him to see phases of Venus, which supported the heliocentric model
- 4 Saw many stars and patterns because it allowed for about a 30x zoom.



Newton's Reflecting Telescope

Replica of Newton's
second telescope

1672



The Newtonian telescope, invented by Isaac Newton in 1668, is a type of reflecting telescope that uses a concave mirror and a flat secondary mirror to eliminate chromatic aberration found in refracting telescopes. It offered a simpler, more affordable design that became widely popular, especially among amateur astronomers. Despite needing occasional alignment and having some optical trade-offs, it revolutionized telescope design.



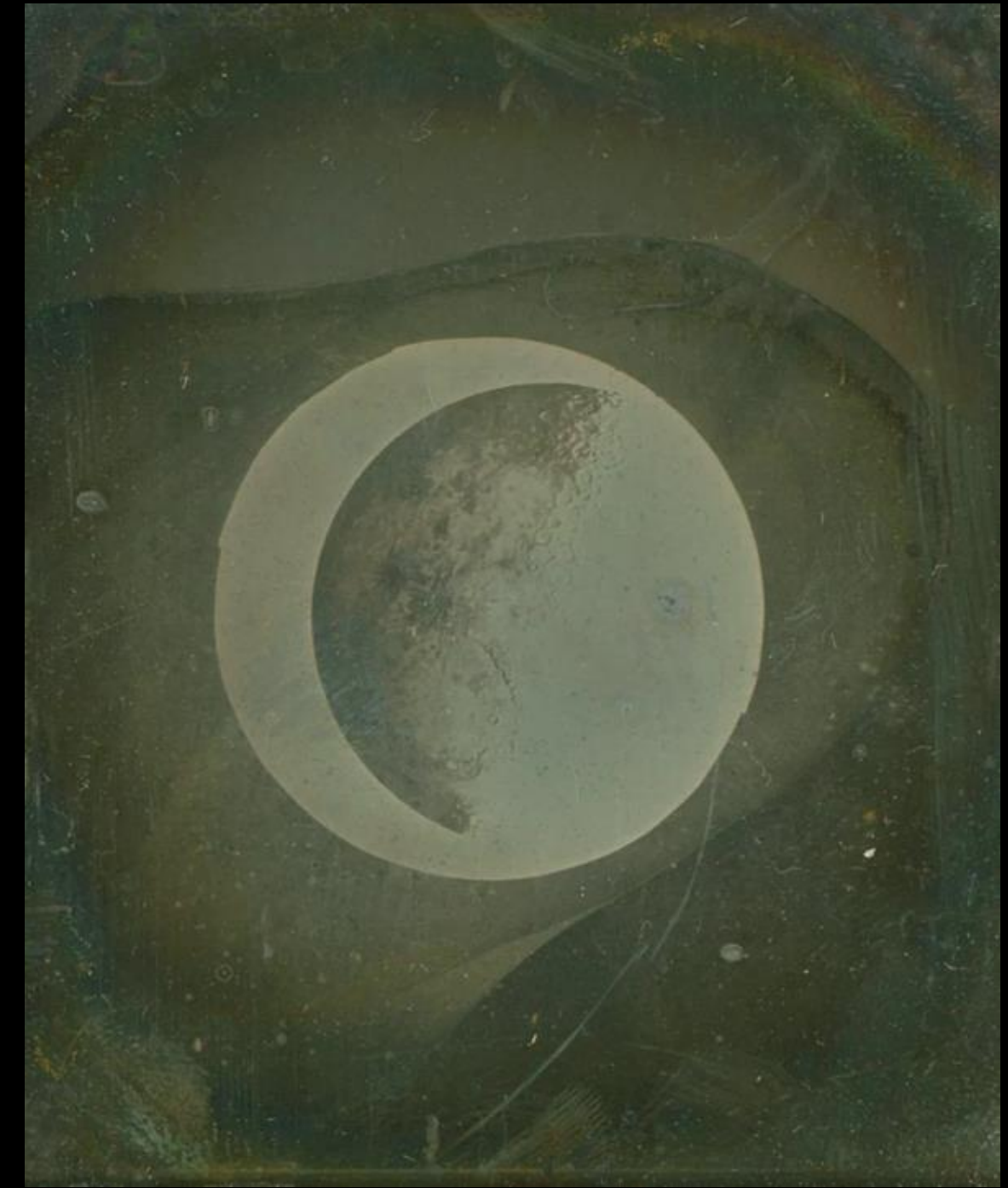
[images](#)

Oldest photo of the Moon



“Louis Dageurre, the inventor of the daguerreotype process, is believed to have taken the very first photo of the Moon in 1839 but his laboratory later burned down destroying his work.

This means that the oldest surviving photo of the Moon was taken by John William Draper from the rooftop observatory at New York University on March 23, 1840. Draper took a 20-minute long exposure using the daguerreotype process and a 5-inch reflecting telescope.”



[images](#)

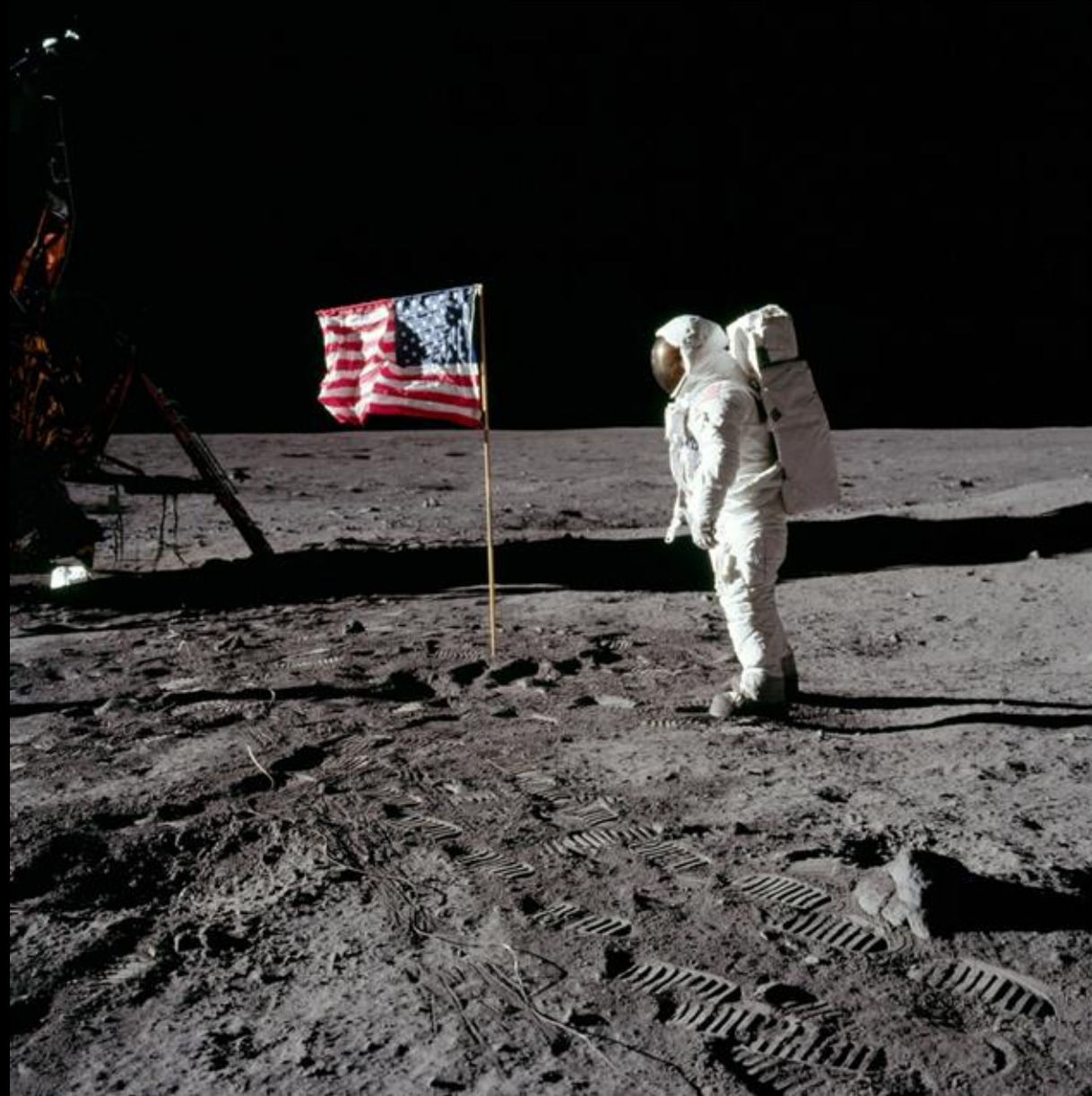
John A. Whipple
James W. Black
1857



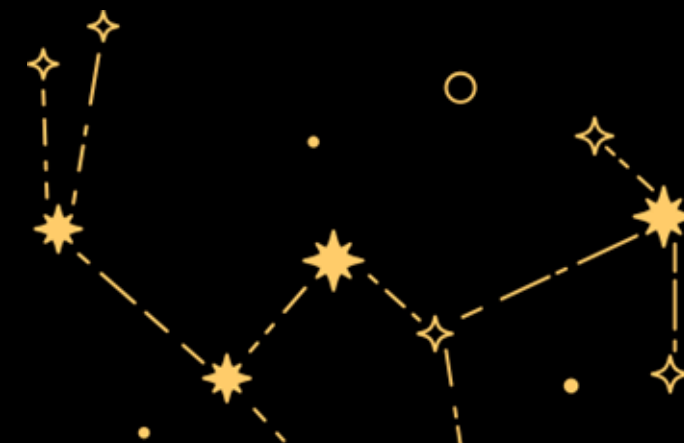
A decorative frame composed of concentric circles and horizontal lines, with constellations and a crescent moon at the corners.

MODERN ERA

Apollo Missions



The Apollo missions successfully landed humans on the Moon for the first time, proving once and for all that space travel beyond Earth orbit was possible. They advanced science and technology, provided valuable lunar data, and symbolized a major achievement in human exploration during the space race.

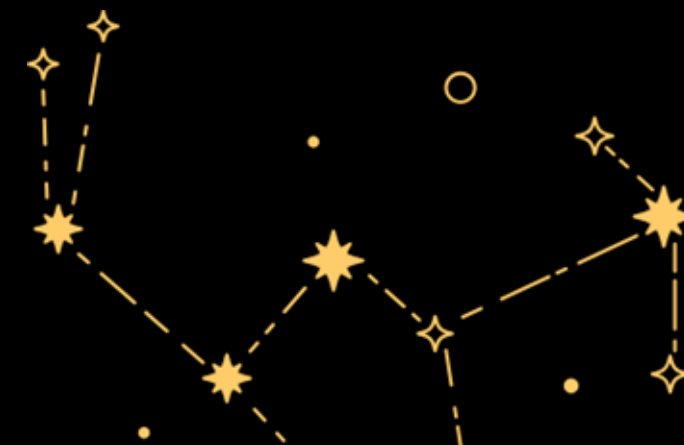


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Hubble Telescope

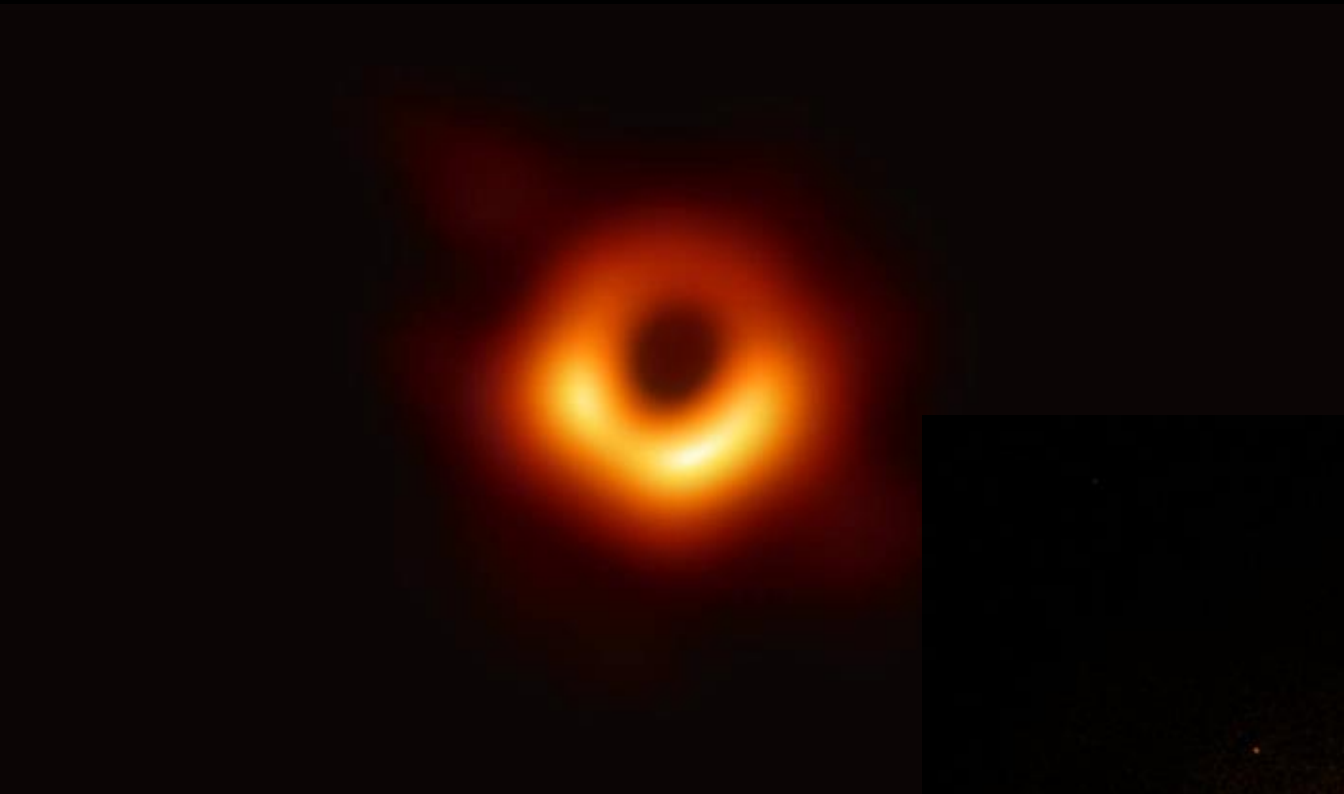


The Hubble Space Telescope was important because it provided the first clear, detailed images of space from outside Earth's atmosphere, revolutionizing our understanding of the universe. It helped determine the rate of the universe's expansion, revealed the existence of dark energy, and captured iconic deep-field images showing thousands of distant galaxies.

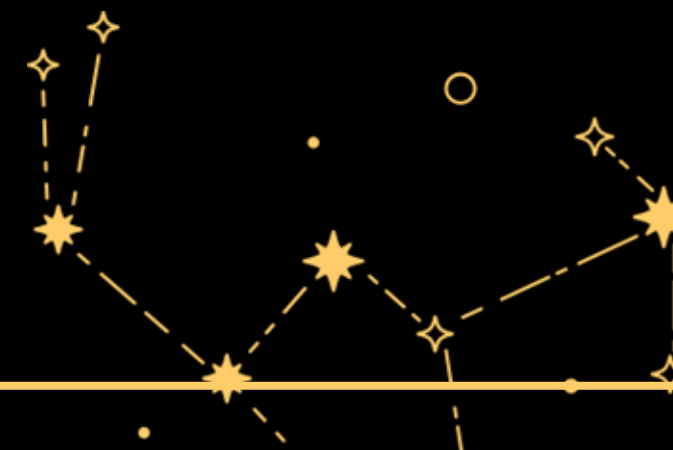
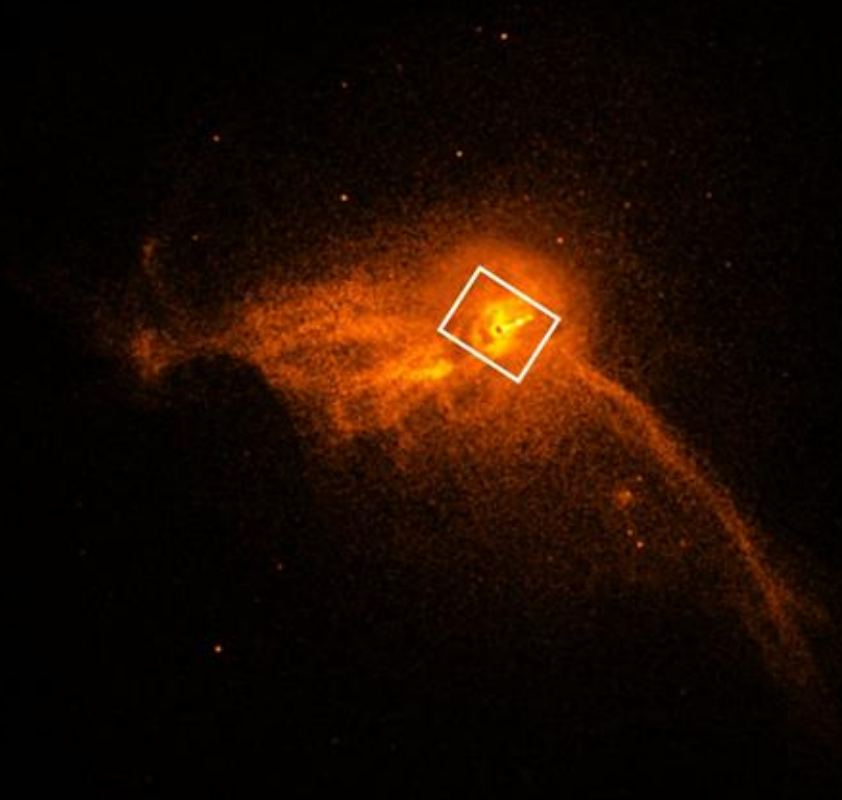


pillars

Event Horizon Telescope (EHT)

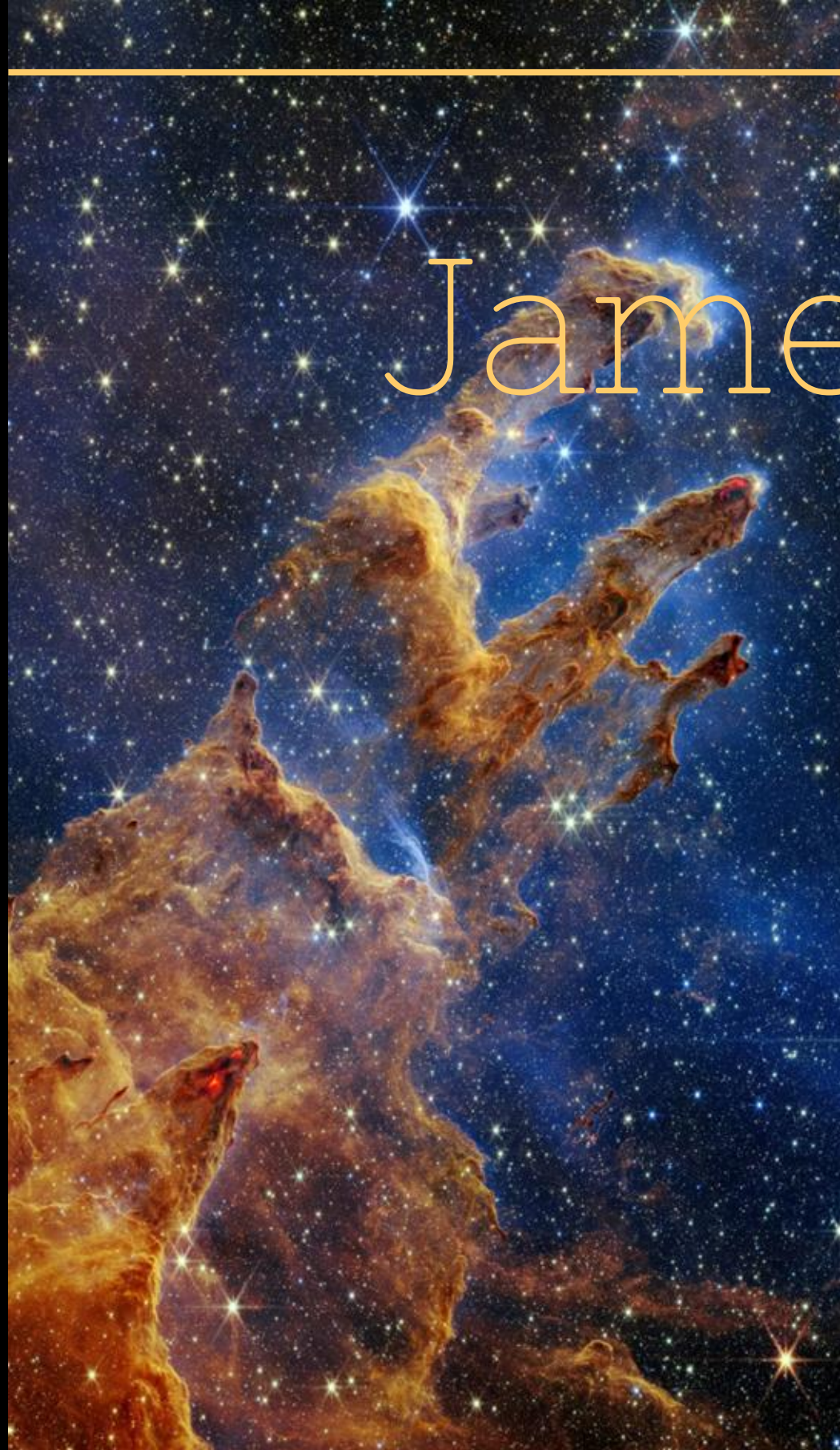


Scientists captured the first image of a black hole by using the EHT, a global network of radio telescopes. By synchronizing these telescopes through a technique called Very Long Baseline Interferometry (VLBI), they effectively created a virtual telescope the size of Earth. This allowed them to image the silhouette of the supermassive black hole at the center of galaxy M87, outlined by the glowing gas surrounding its event horizon. Public release 2019

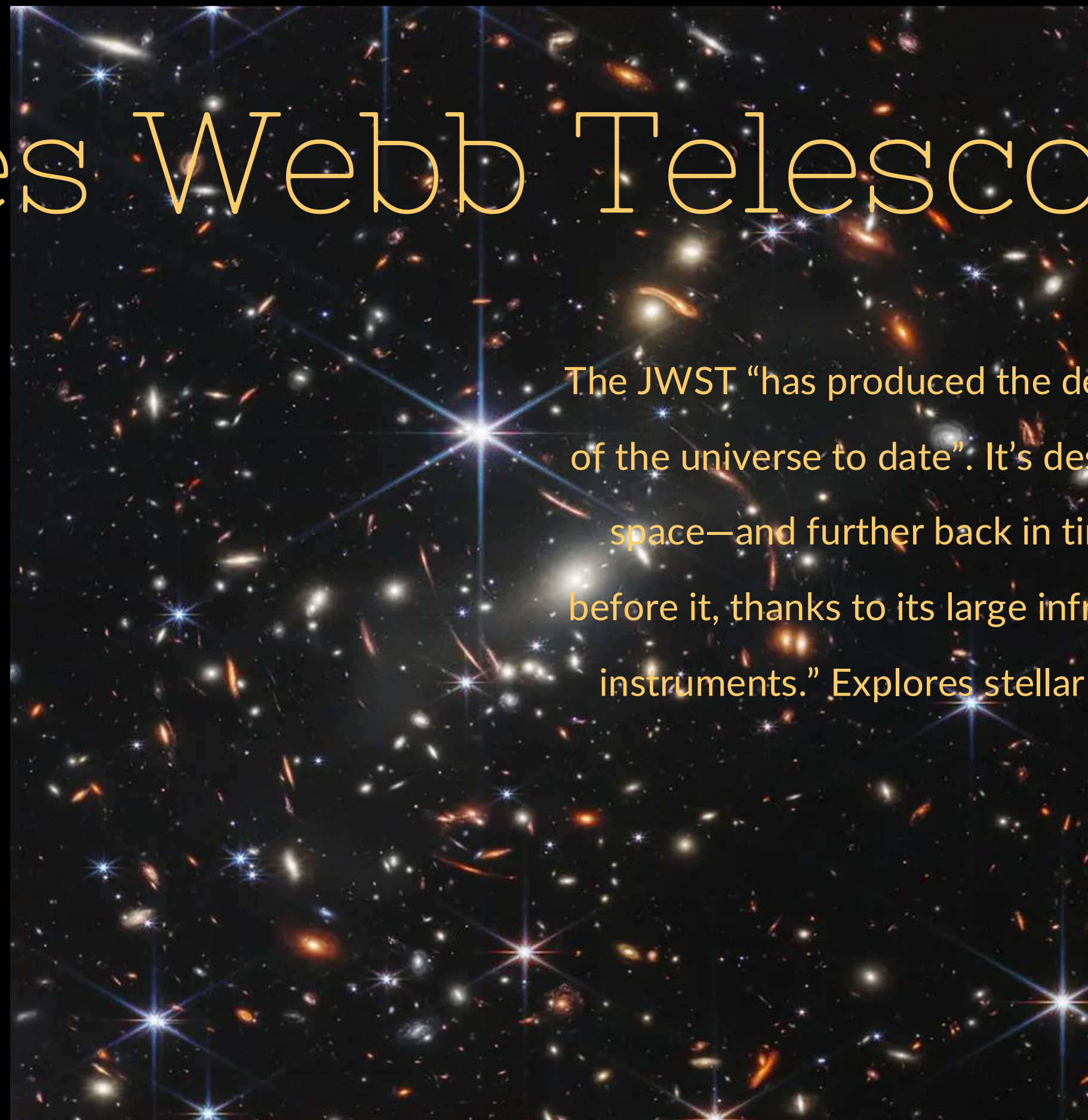


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James Webb Telescope



Stretching roughly 4 to 5 light-years, the Pillars of Creation are a fascinating but relatively small feature of the entire Eagle Nebula, which spans 70 by 55 light-years.



The JWST “has produced the deepest and sharpest image of the universe to date”. It’s designed to look deeper into space—and further back in time—than any telescope before it, thanks to its large infrared mirror and advanced instruments.” Explores stellar life cycles with infrared.

[mini article](#)

[best pics](#)

Links Used:

- <https://www.businessinsider.com/ancient-cave-drawings-are-constellations-of-stars-2018-12>
- https://archive.org/details/cambridgeillustr0000unse_q5p9/page/22/mode/2up
- https://www.britishmuseum.org/collection/object/W_1863-0826-1
- https://en.wikipedia.org/wiki/Babylonian_star_catalogues
- <https://mhs.web.ox.ac.uk/collections-online#/item/hsm-catalogue-2110>
- <https://en.wikipedia.org/wiki/Astrolabe>
- <https://ecuip.lib.uchicago.edu/multiwavelength-astronomy/optical/history/03.html>
- <https://www.britannica.com/science/optical-telescope>
- https://en.wikipedia.org/wiki/Newtonian_telescope
- <https://petapixel.com/2023/06/14/a-brief-history-of-the-very-first-moon-photos-ever-taken/>
- <https://www.nasa.gov/image-article/nasas-webb-delivers-deepest-infrared-image-of-universe-yet/>
- <https://fstoppers.com/originals/first-photograph-taken-moon-neil-armstrong-featured-trash-bag-243698>

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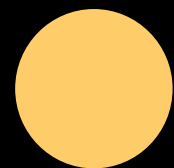
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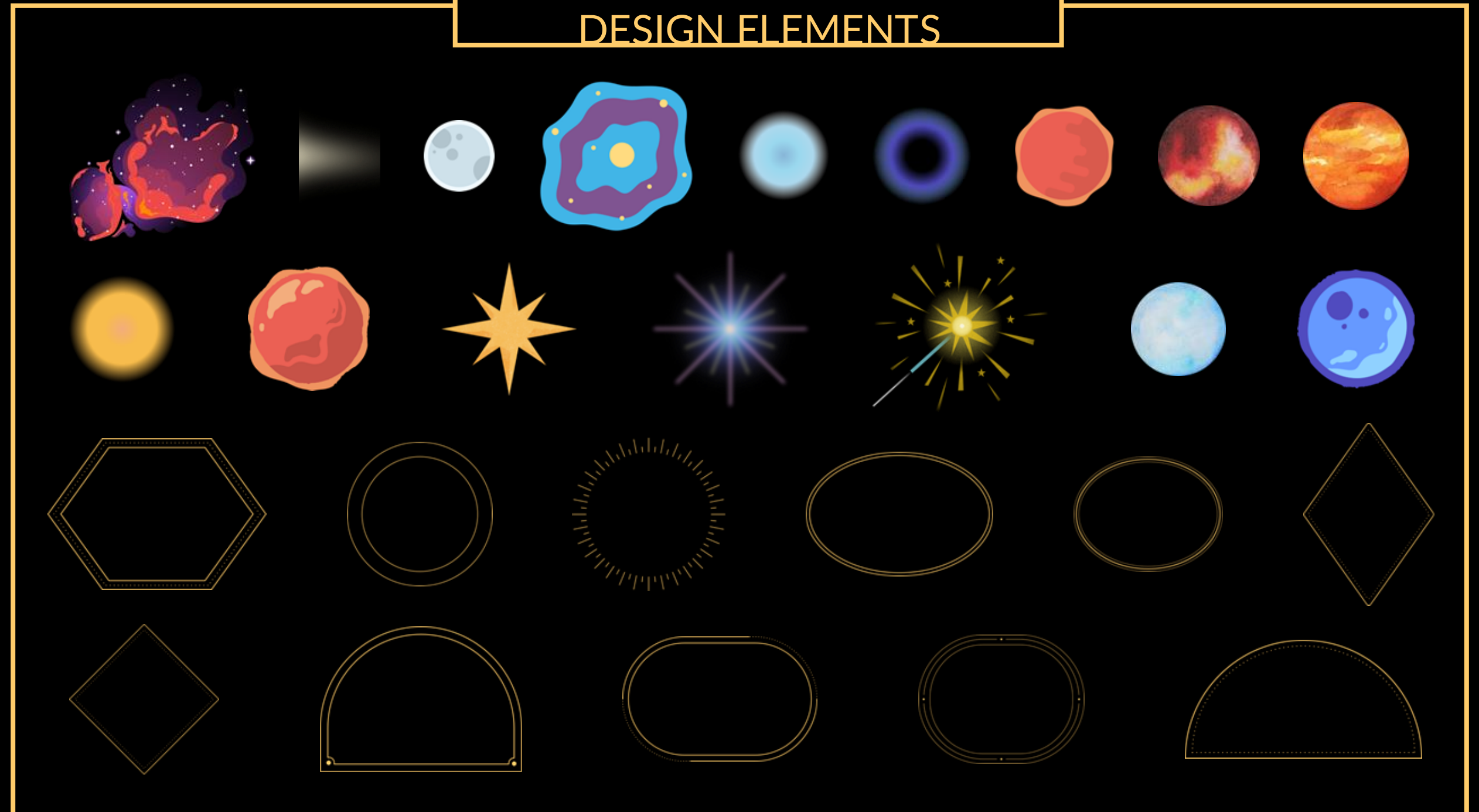


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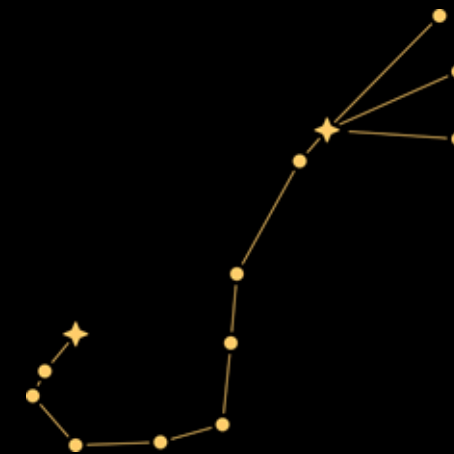
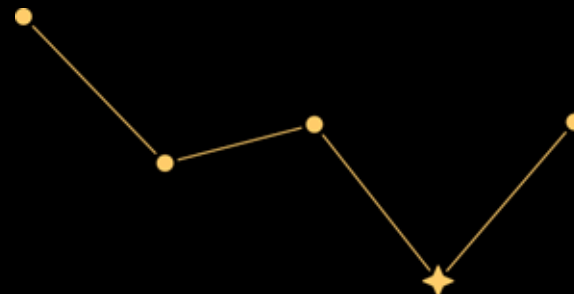
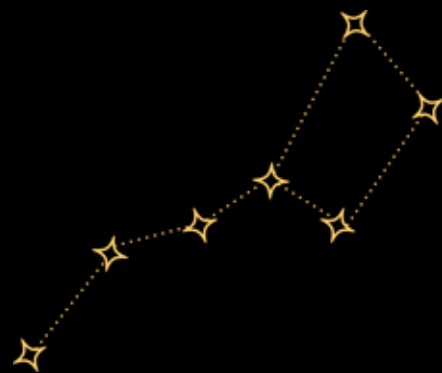
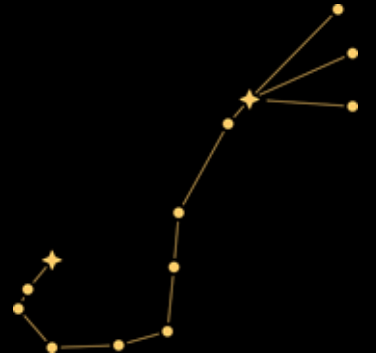
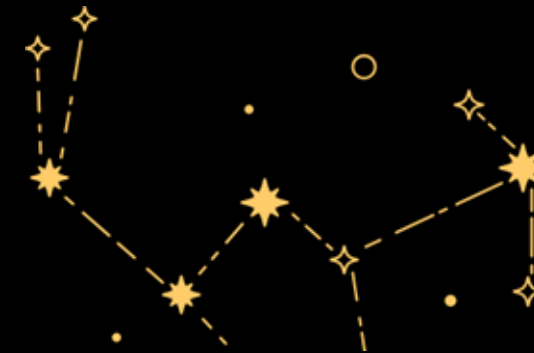
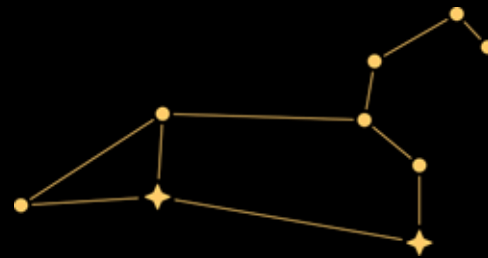
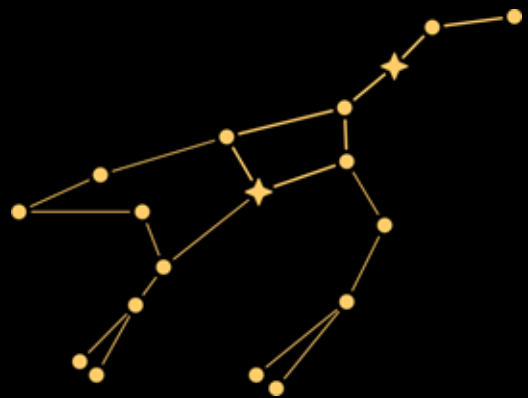


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DESIGN ELEMENTS



DESIGN ELEMENTS



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